

Benchmark notes

- LAK is safe-Rust.
- All benchmarks are single-threaded on an Apple M4 with 16 GB unified memory.
- Comparisons use OpenBLAS armv8, BLIS, and Apple Accelerate where shown.
- BLIS is omitted from Level-1 and Level-2 because this backend is not optimized for those routines.
- Accelerate is omitted from Level-2 and Level-3 because it uses Apple's AMX which is cheating.
- OpenBLAS armv8 is included throughout; but its ARM64 kernels target [Level-2](#) and [Level-3](#) most directly.
- Level-1 and Level-2 report GB/s; Level-3 reports GFLOP/s.
- All matrices are square $n \times n$.

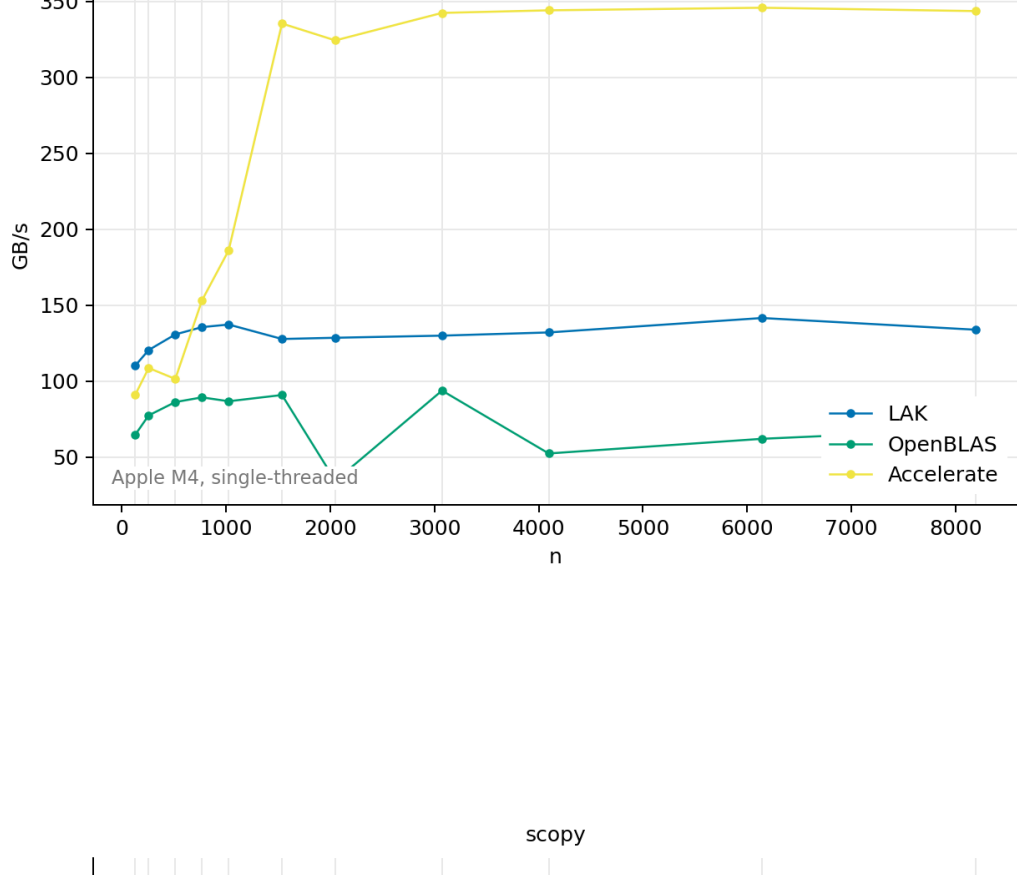
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Level-1

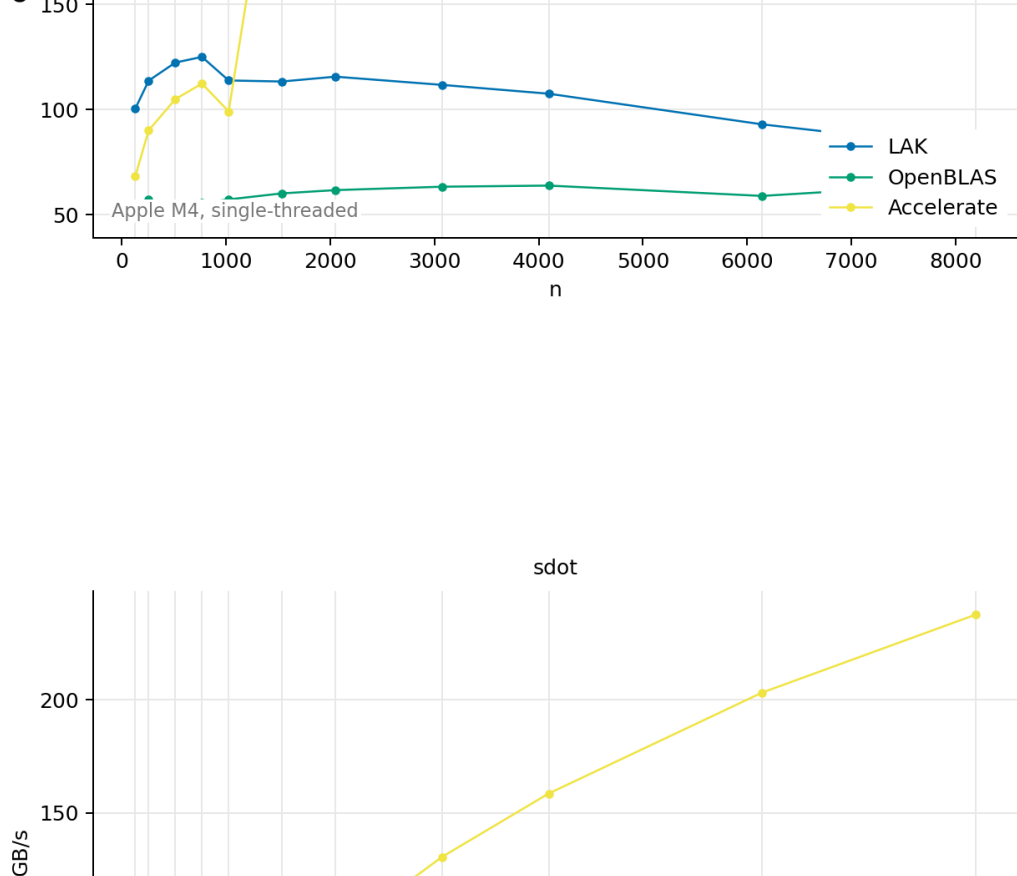
SASUM

$SASUM: s \leftarrow \sum_i |x_i|$



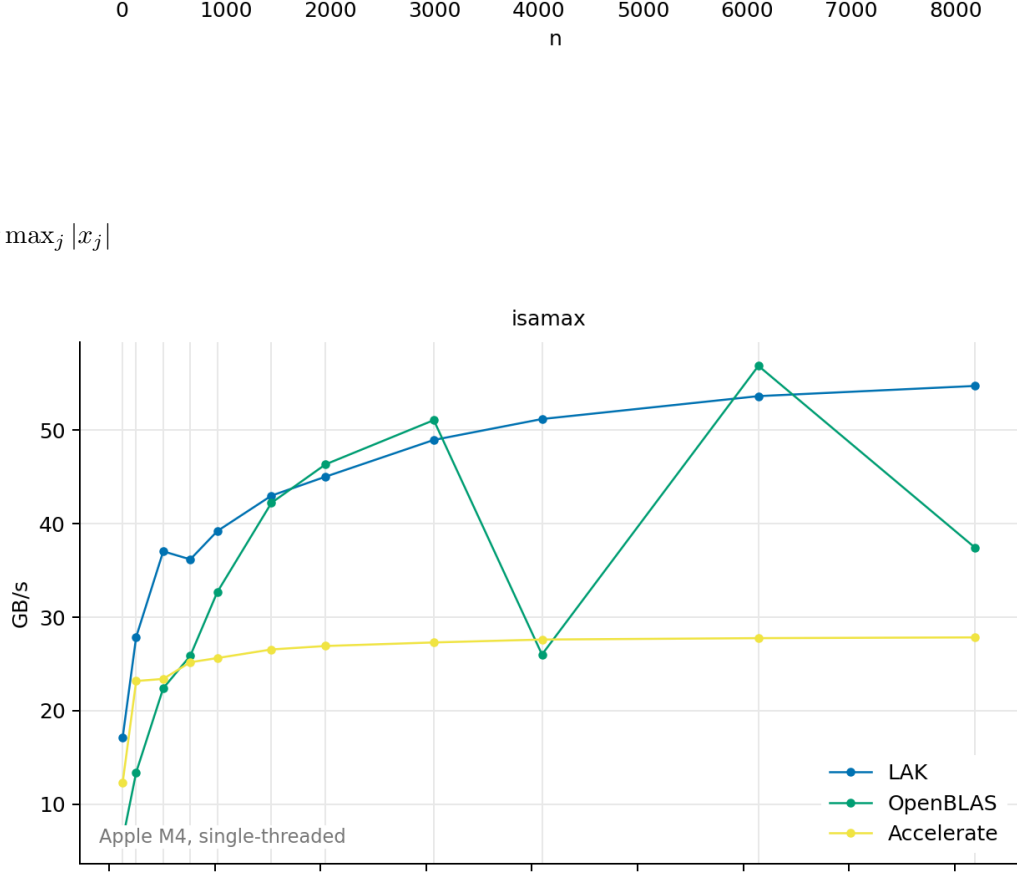
SAXPY

$SAXPY: y \leftarrow \alpha x + y$



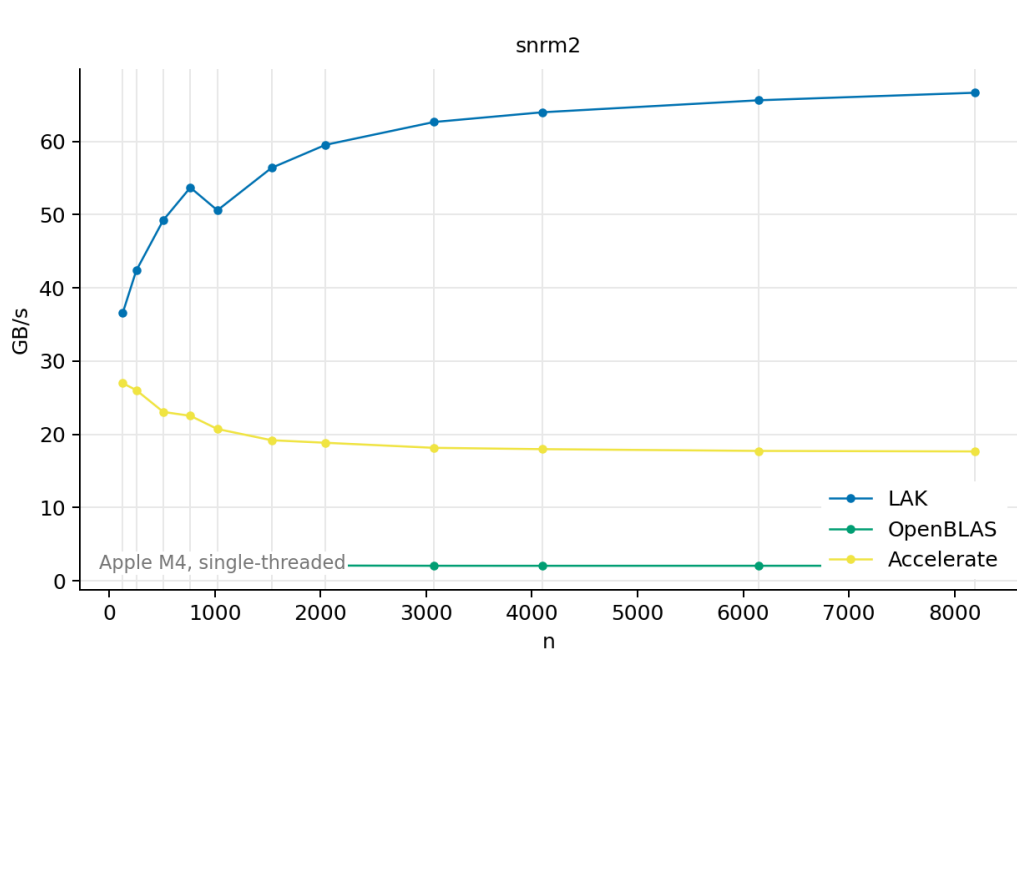
SCOPY

$SCOPY: y \leftarrow x$



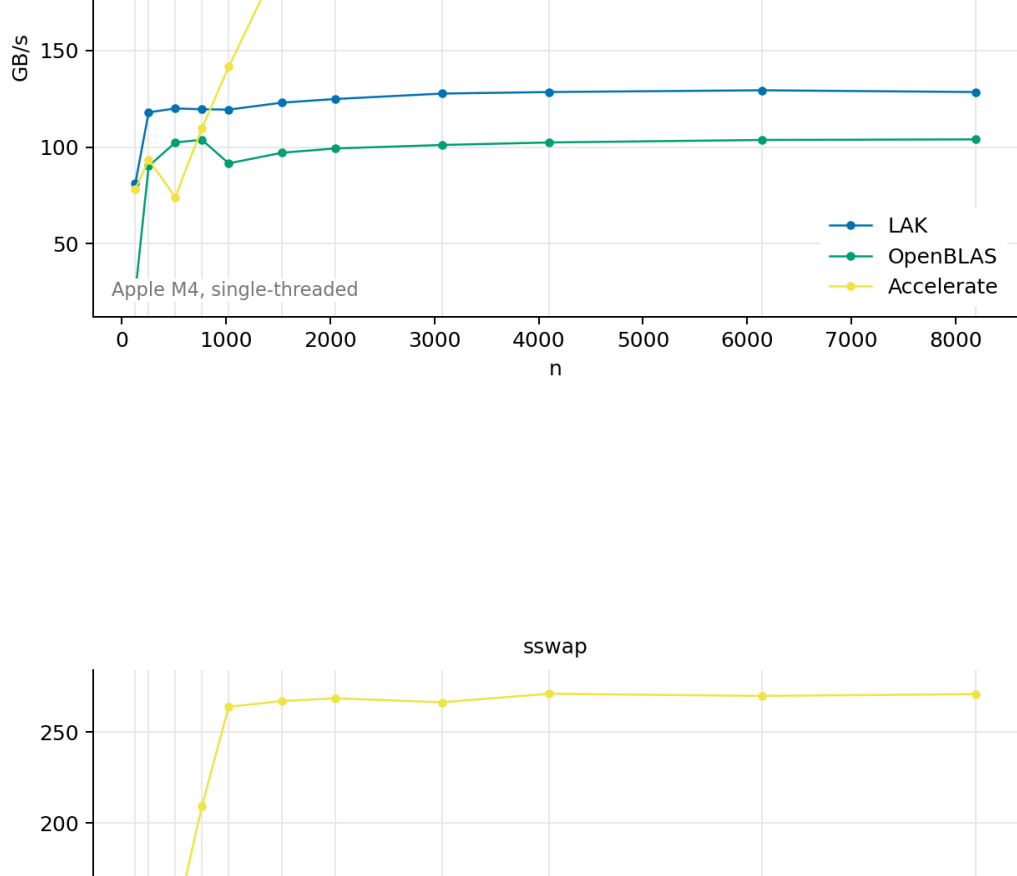
SDOT

$SDOT: s \leftarrow x^T y$



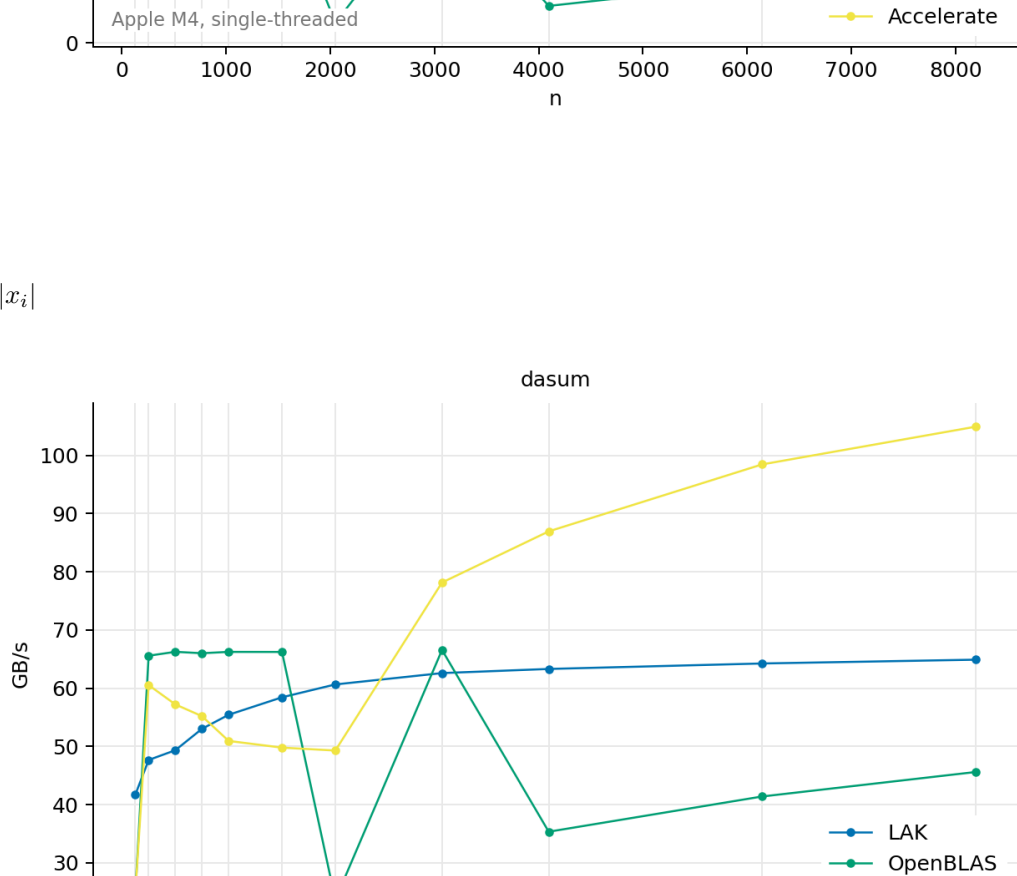
ISAMAX

$ISAMAX: i \leftarrow \arg \max_j |x_j|$



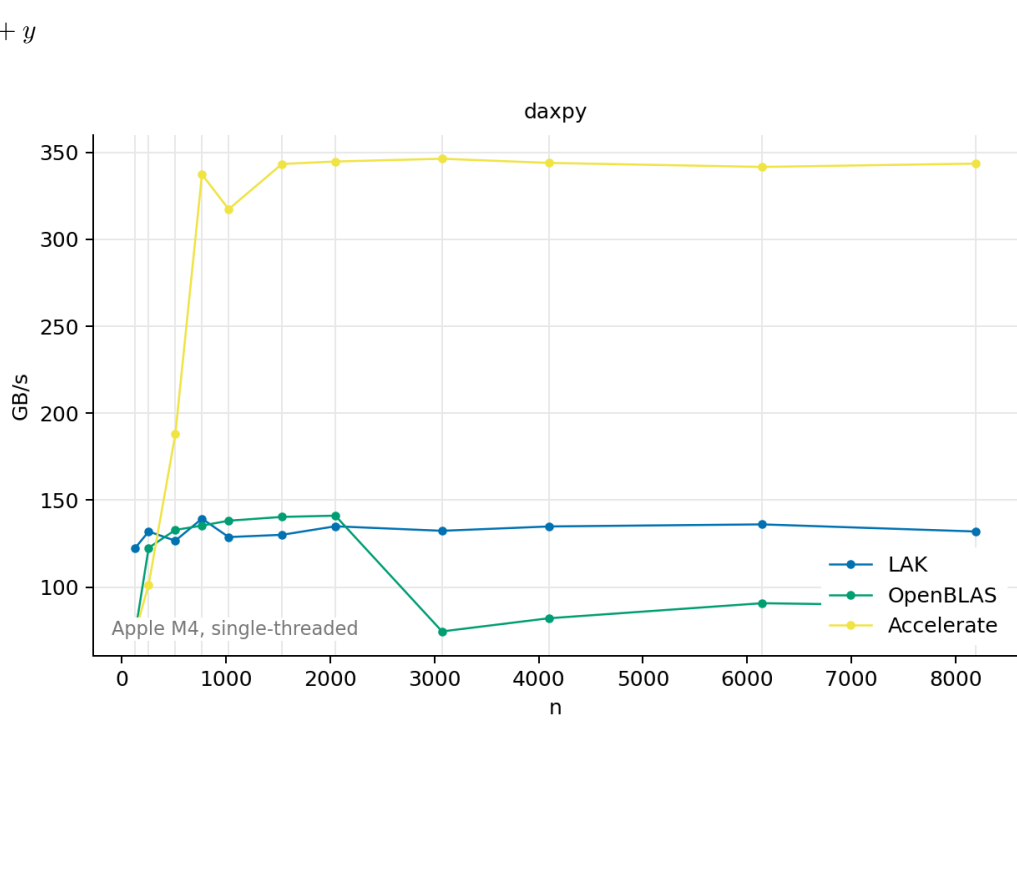
SNRM2

$SNRM2: s \leftarrow \sqrt{x^T x}$



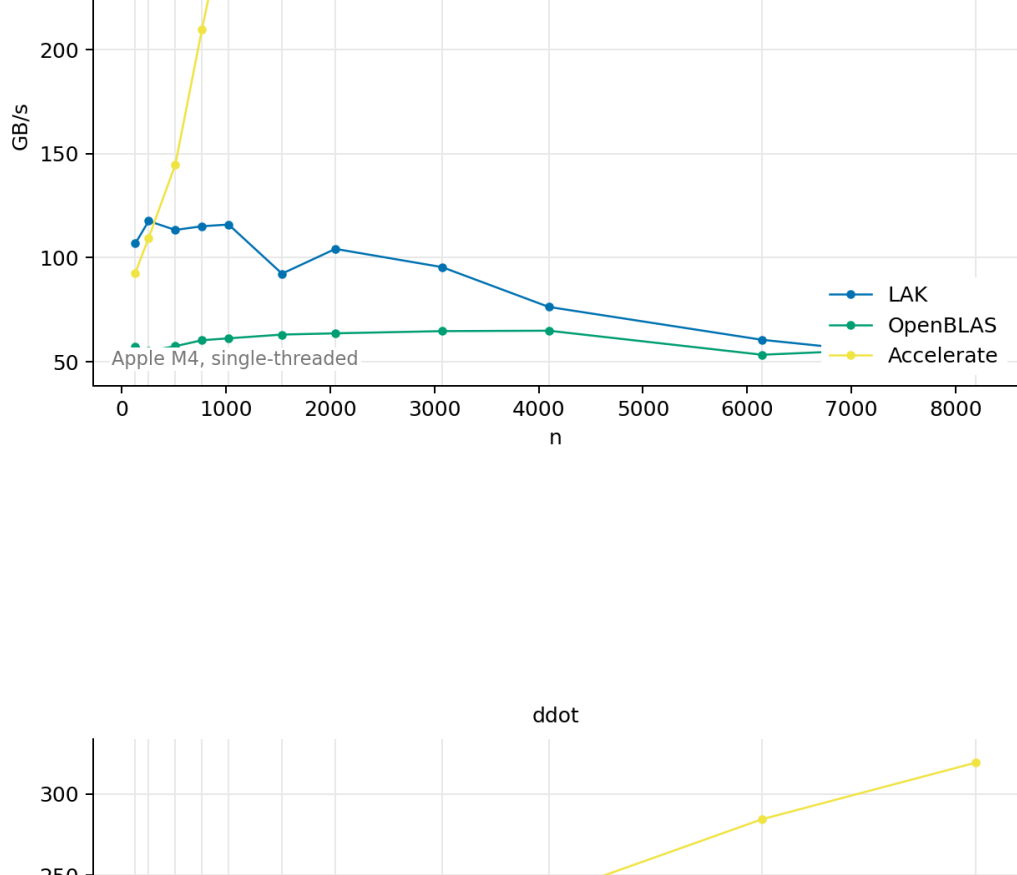
SSCAL

$SSCAL: x \leftarrow \alpha x$



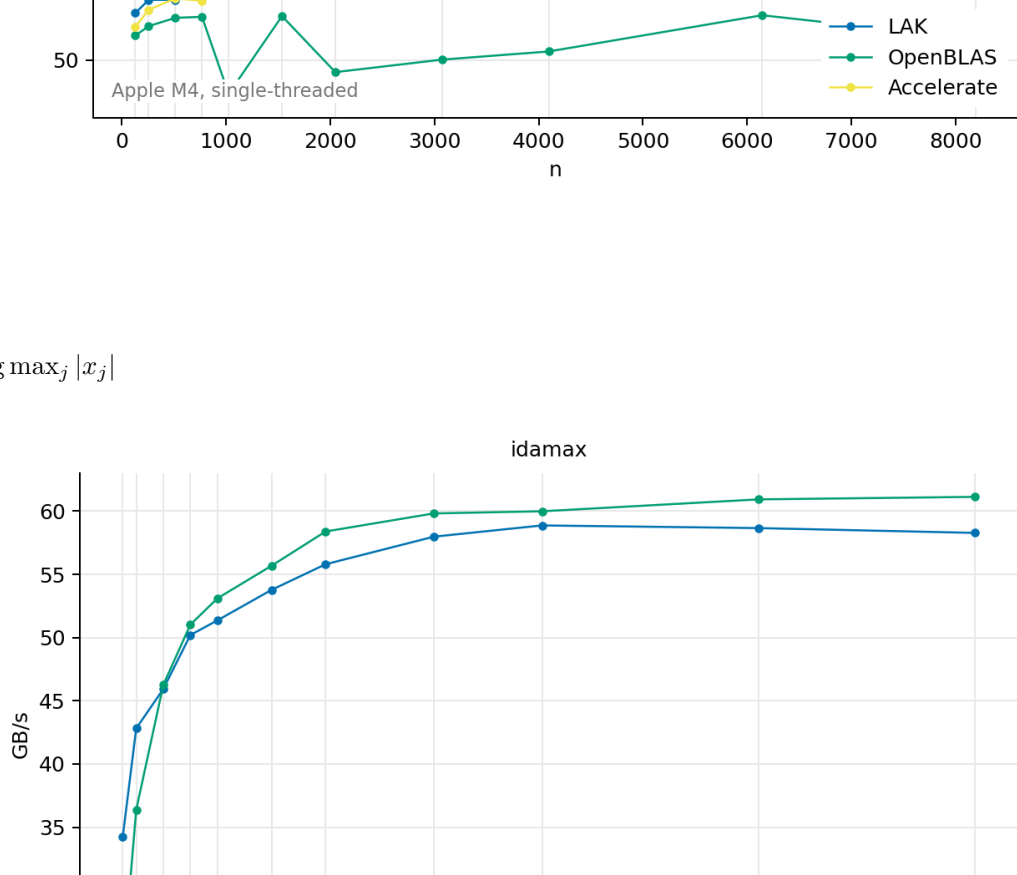
SSWAP

$SSWAP: x \leftrightarrow y$



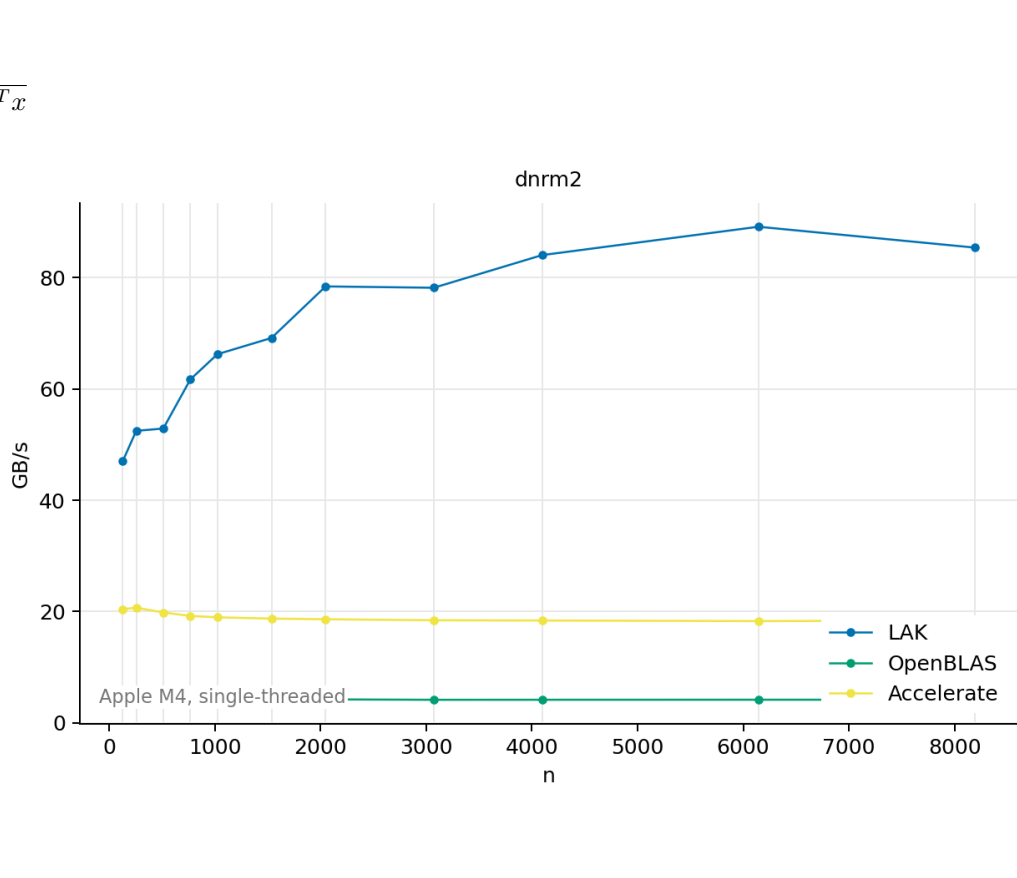
DASUM

$DASUM: s \leftarrow \sum_i |x_i|$



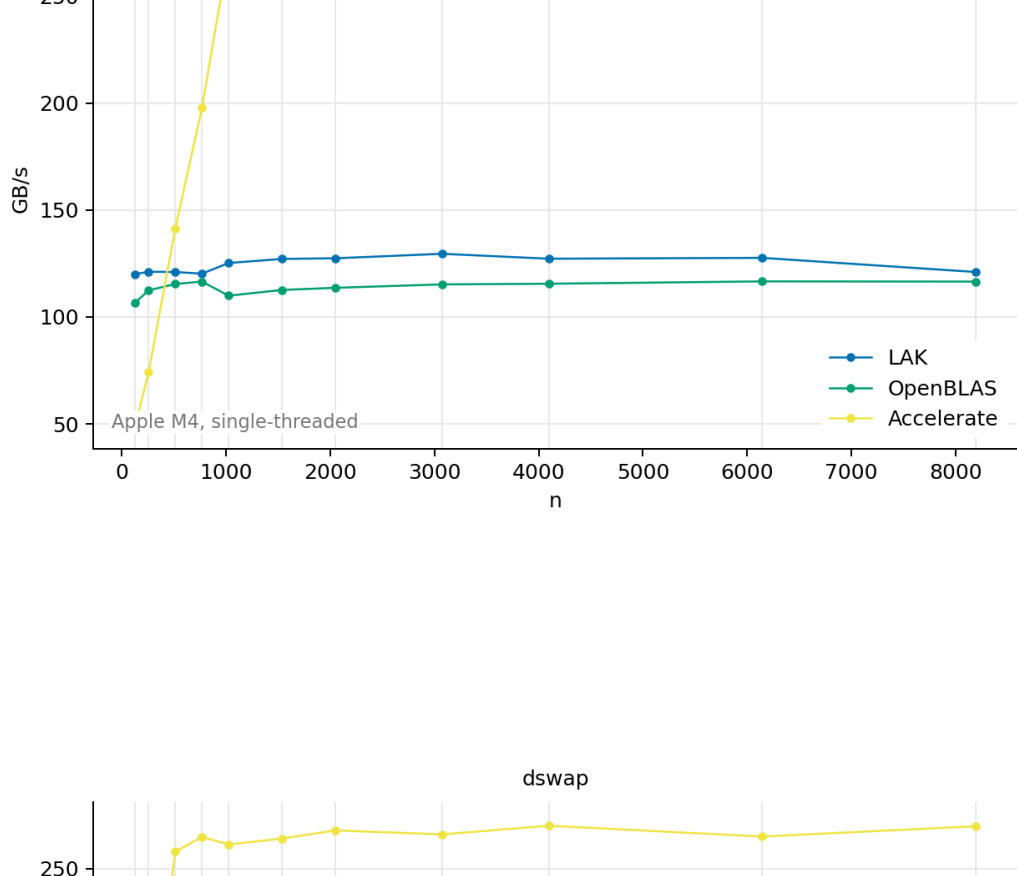
DAXPY

$DAXPY: y \leftarrow \alpha x + y$



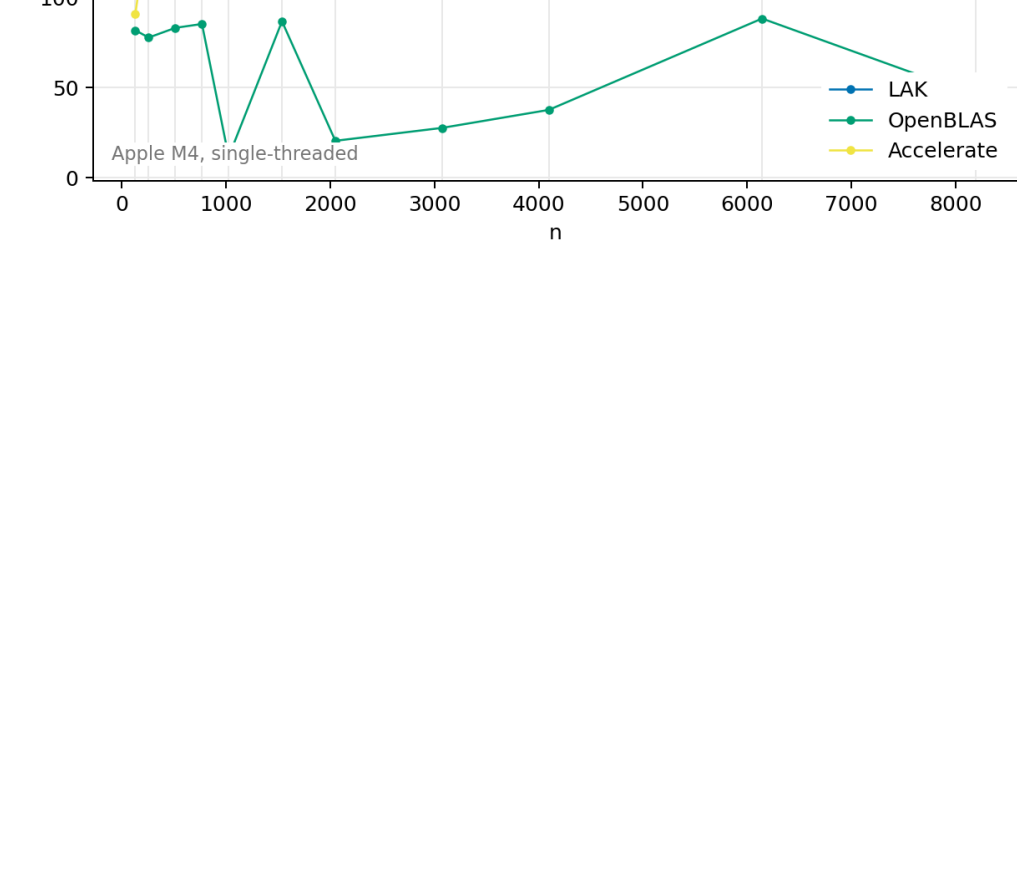
DCOPY

$DCOPY: y \leftarrow x$



DDOT

$DDOT: s \leftarrow x^T y$



IDAMAX

$IDAMAX: i \leftarrow \arg \max_j |x_j|$



DNRM2

$DNRM2: s \leftarrow \sqrt{x^T x}$

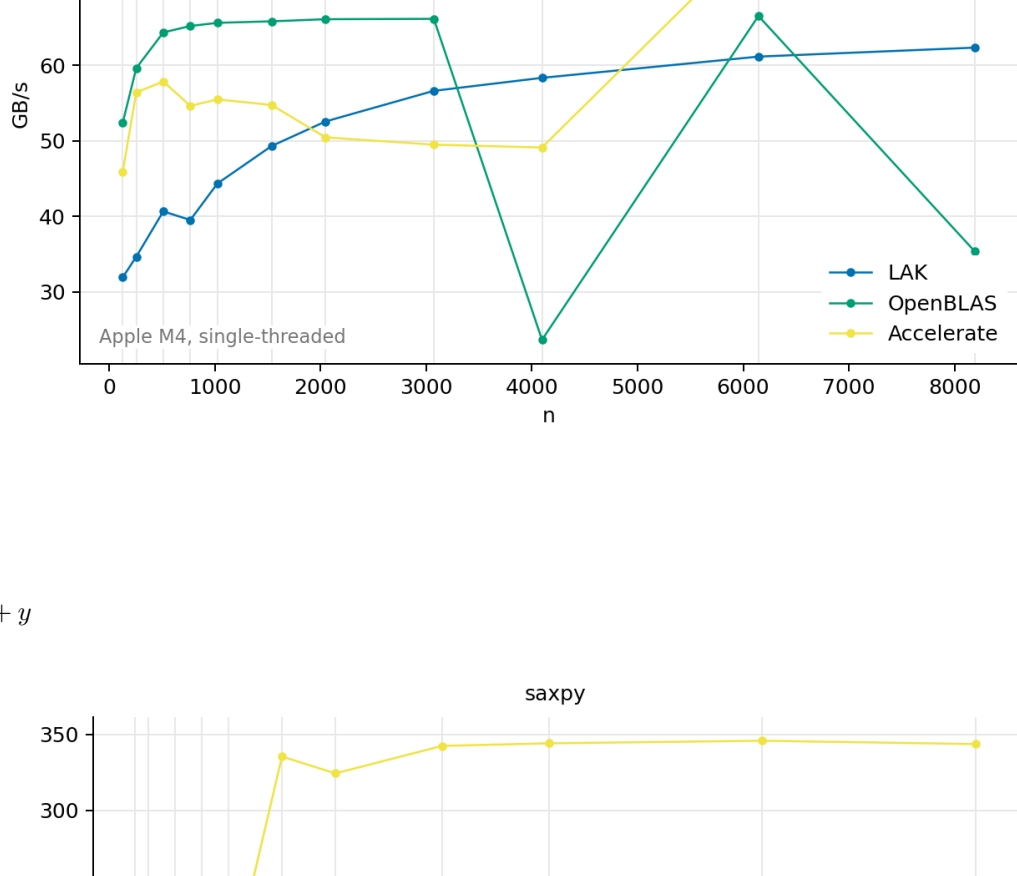


DSCAL

$DSCAL: x \leftarrow \alpha x$

DSWAP

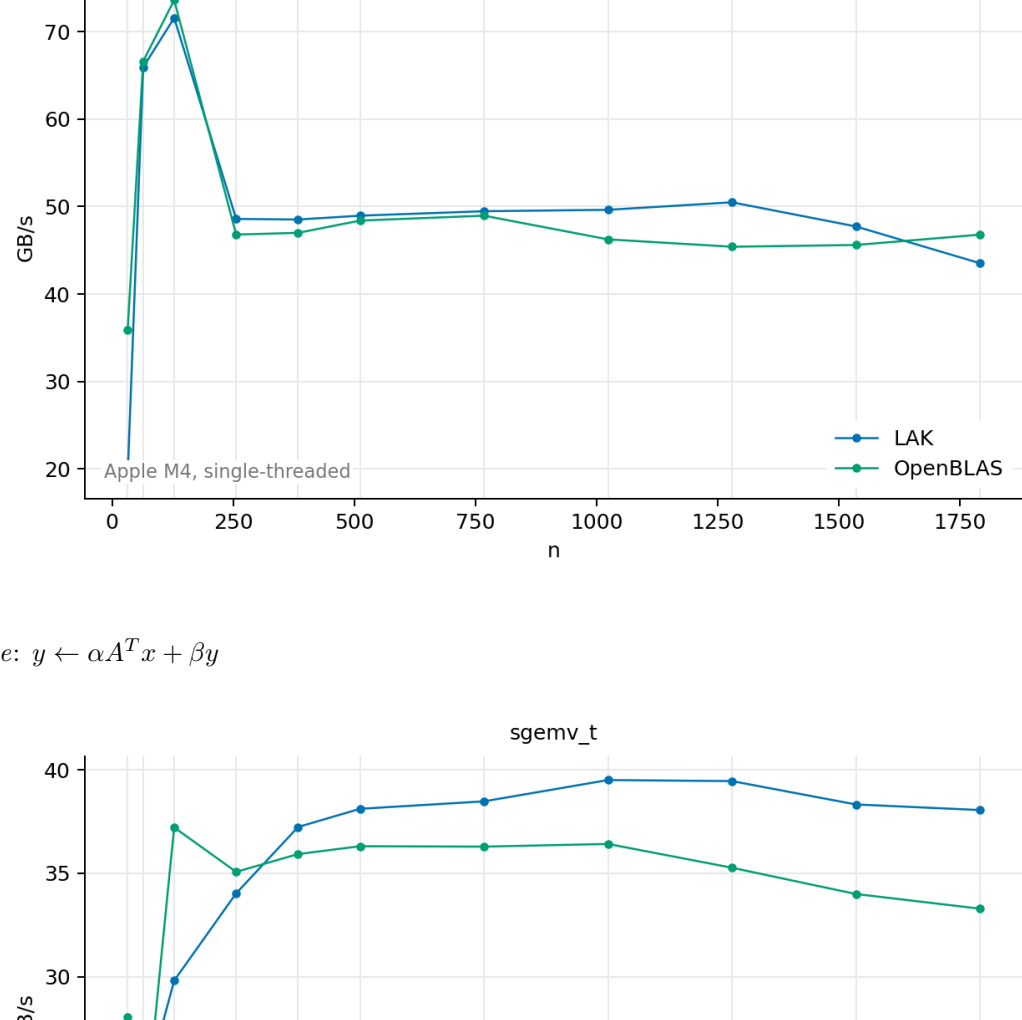
$DSWAP: x \leftrightarrow y$



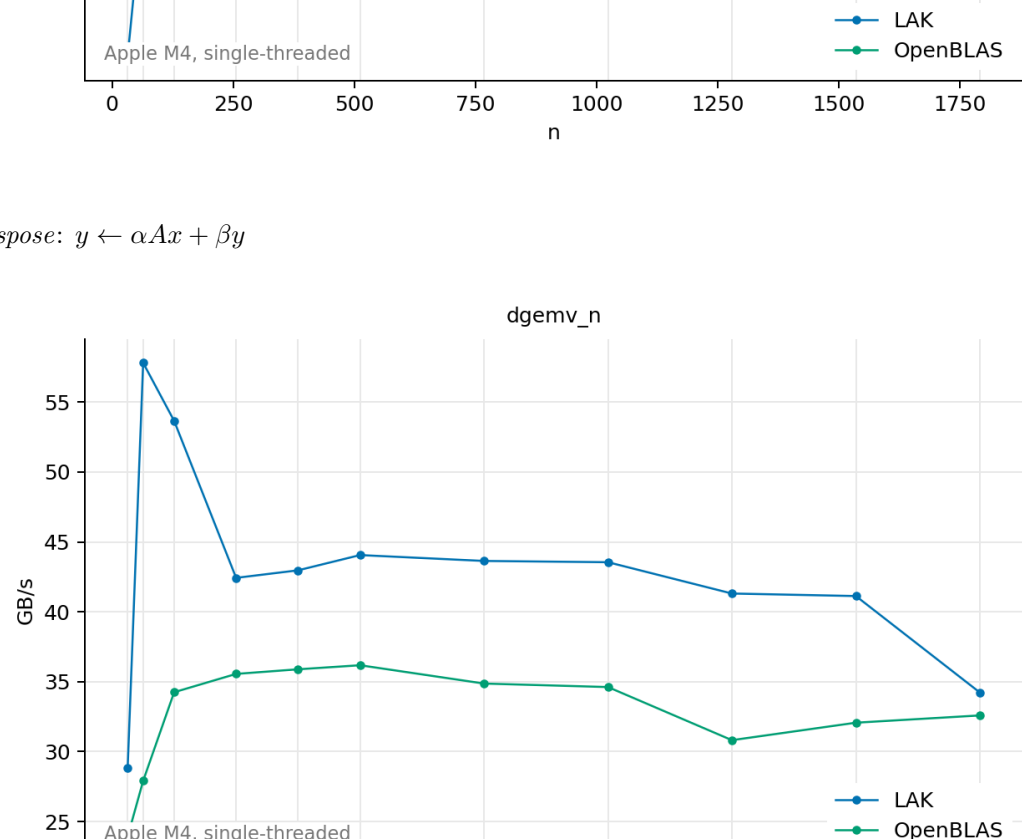
Level-2

GEMV

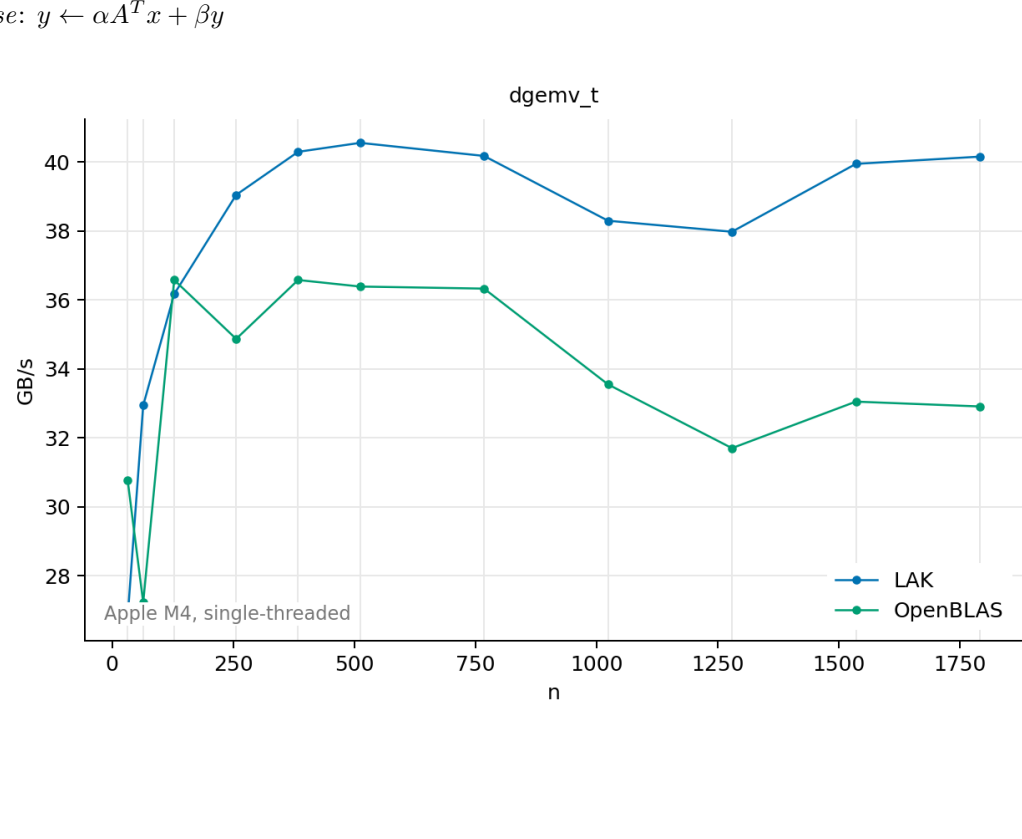
SGEMV, no transpose: $y \leftarrow \alpha Ax + \beta y$



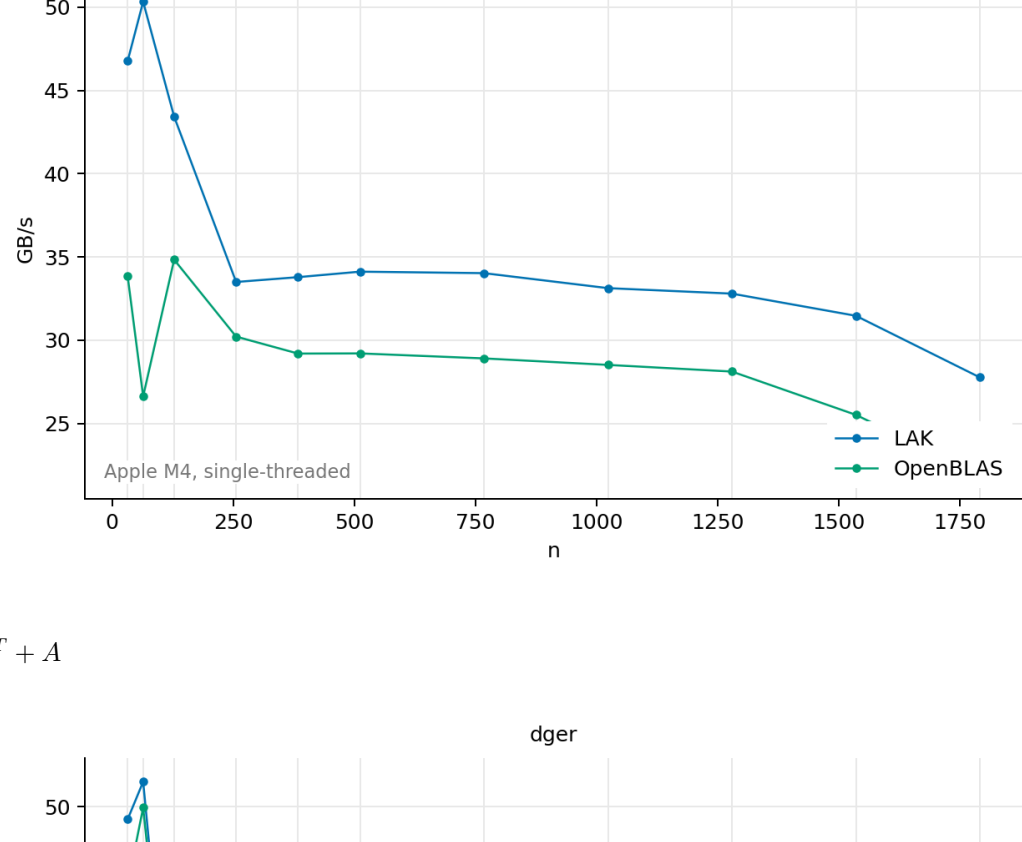
SGEMV, transpose: $y \leftarrow \alpha A^T x + \beta y$



DGEMV, no transpose: $y \leftarrow \alpha Ax + \beta y$

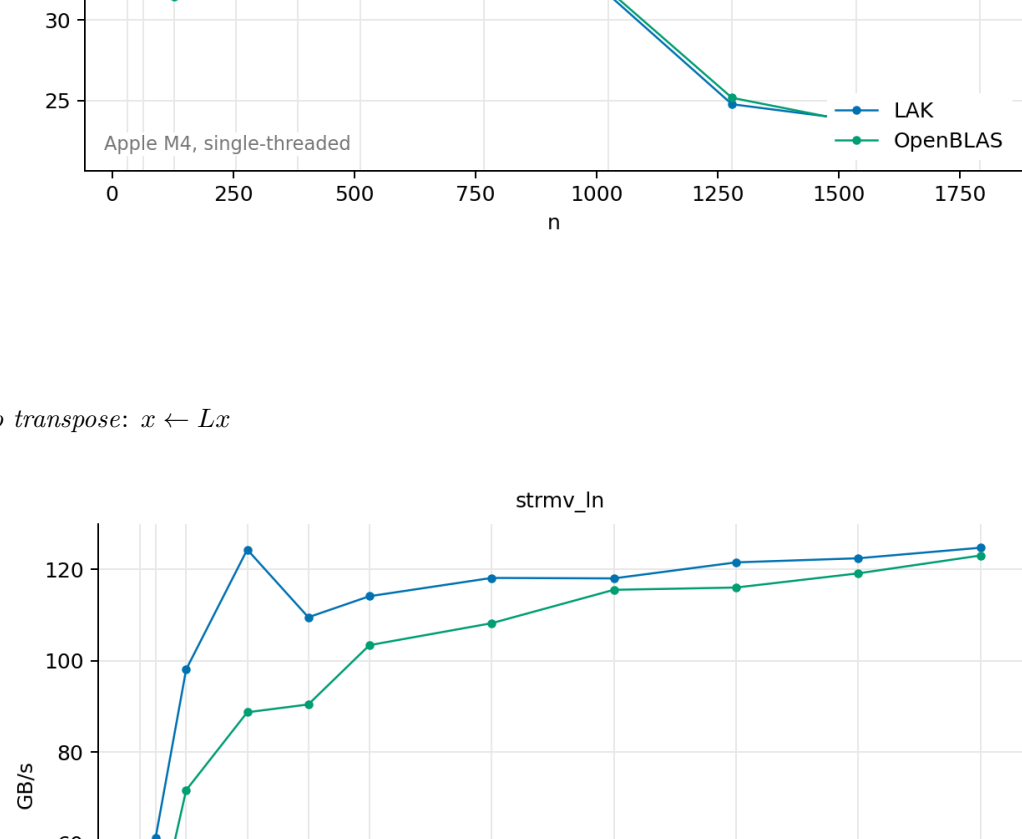


DGEMV, transpose: $y \leftarrow \alpha A^T x + \beta y$

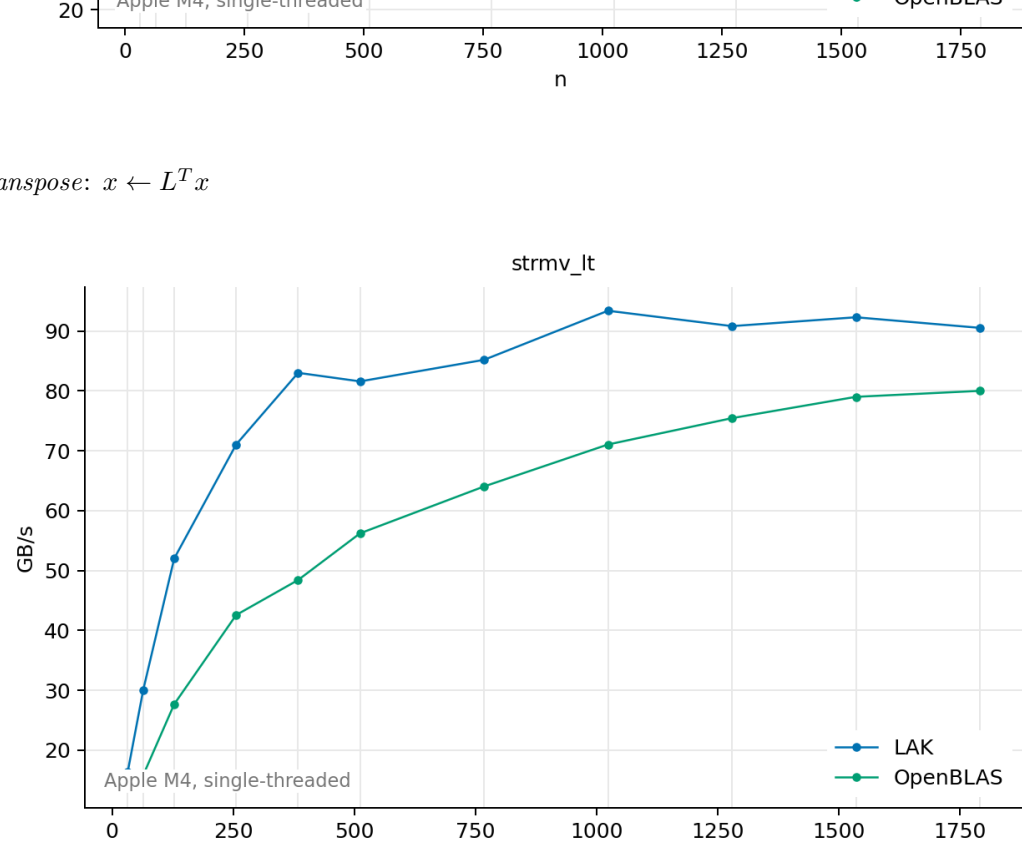


GER

SGER: $A \leftarrow \alpha xy^T + A$

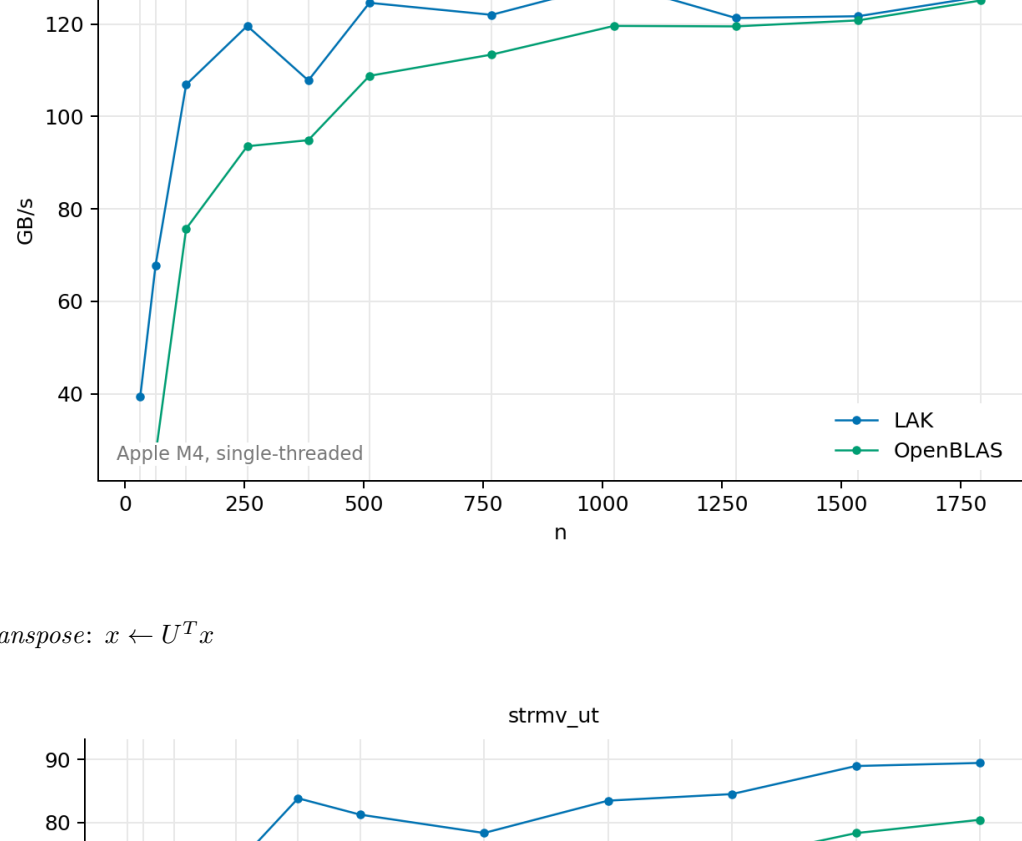


DGER: $A \leftarrow \alpha xy^T + A$

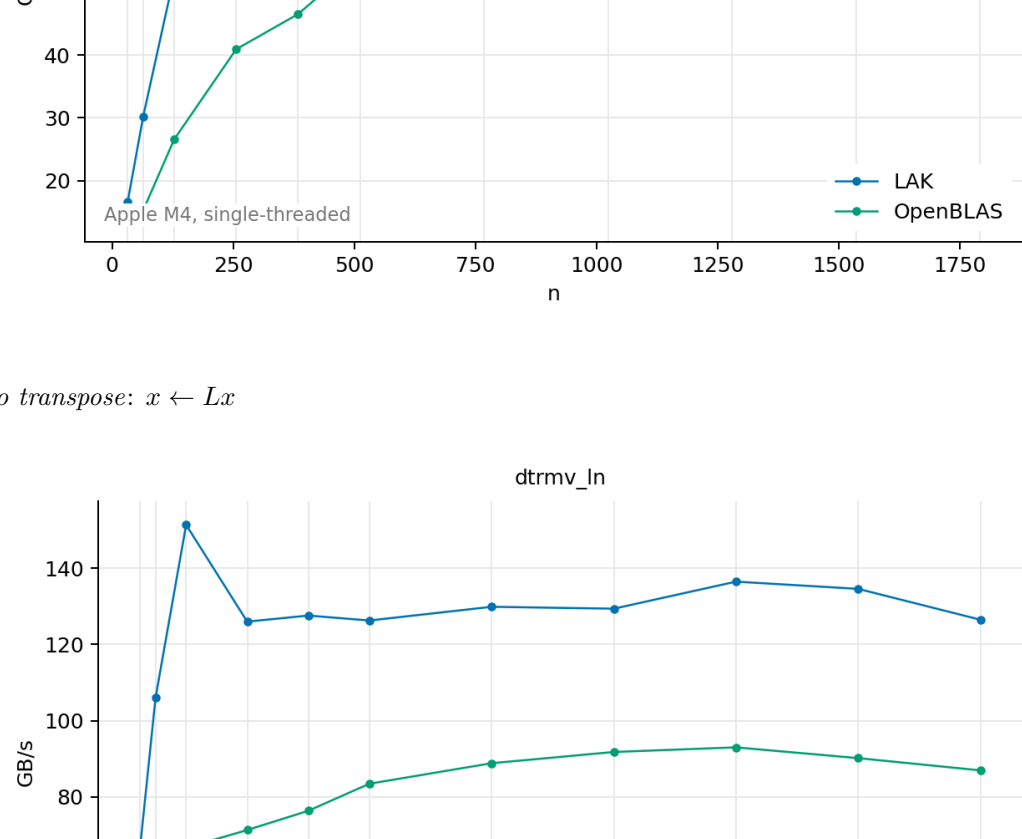


TRMV

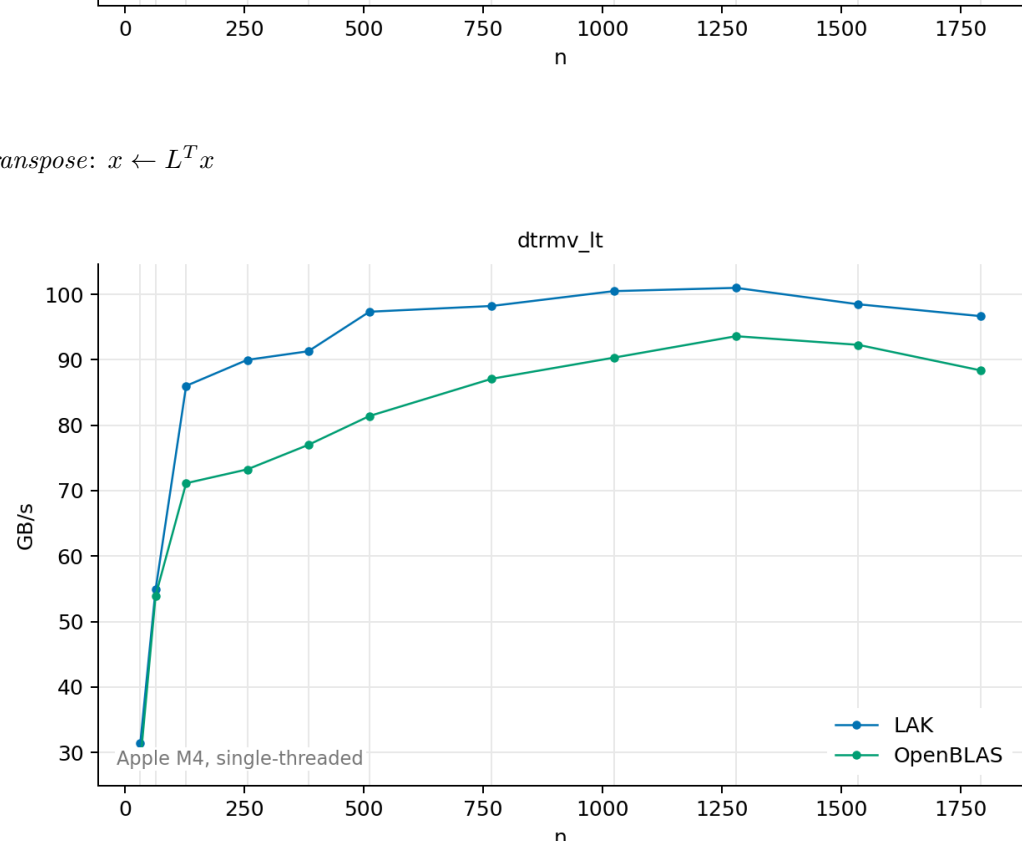
STRMV, lower no transpose: $x \leftarrow Lx$



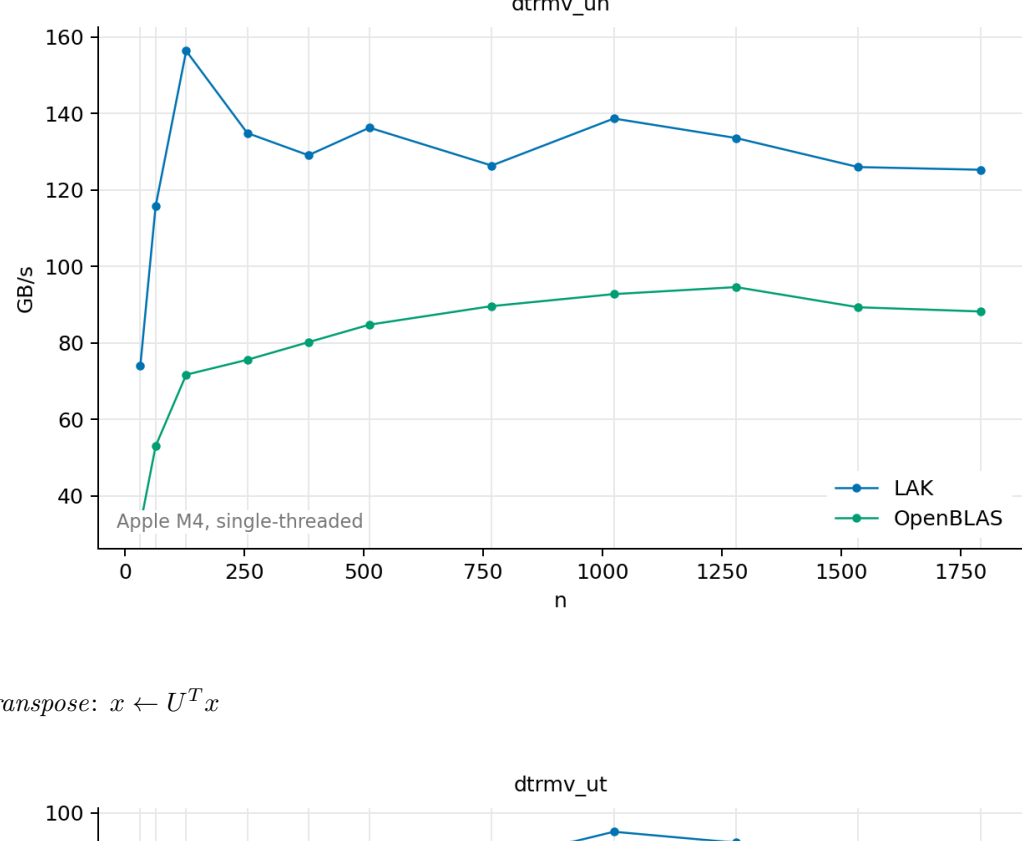
STRMV, lower transpose: $x \leftarrow L^T x$



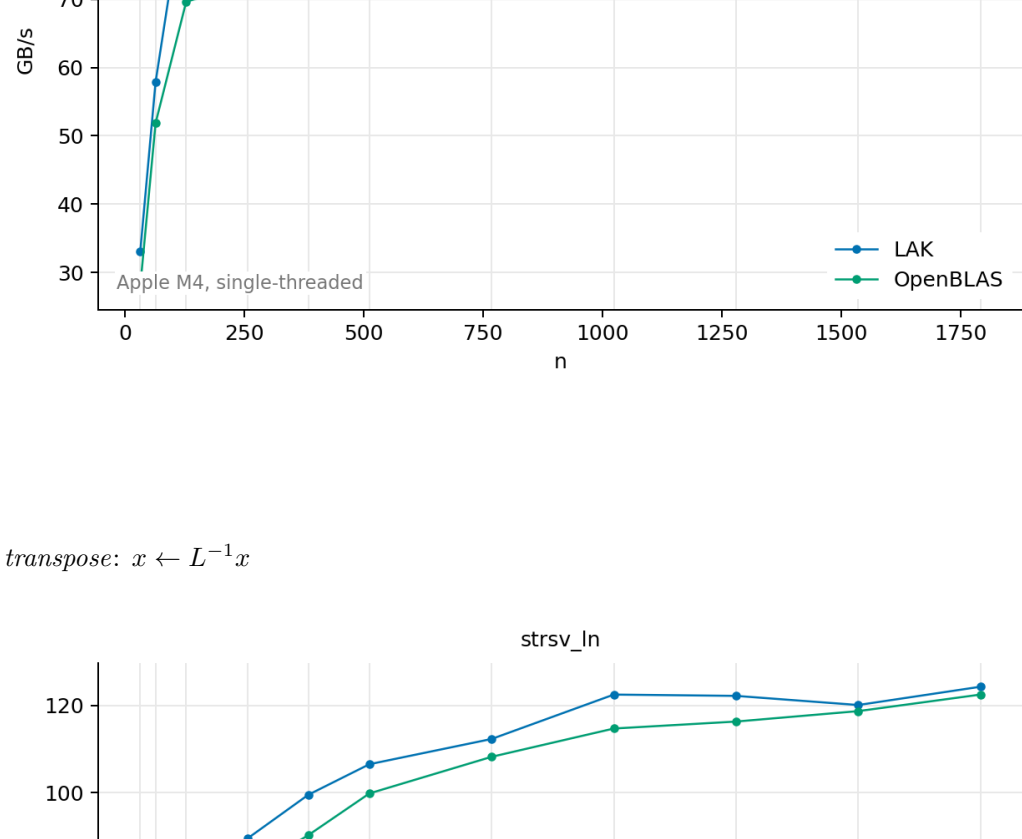
STRMV, upper no transpose: $x \leftarrow Ux$



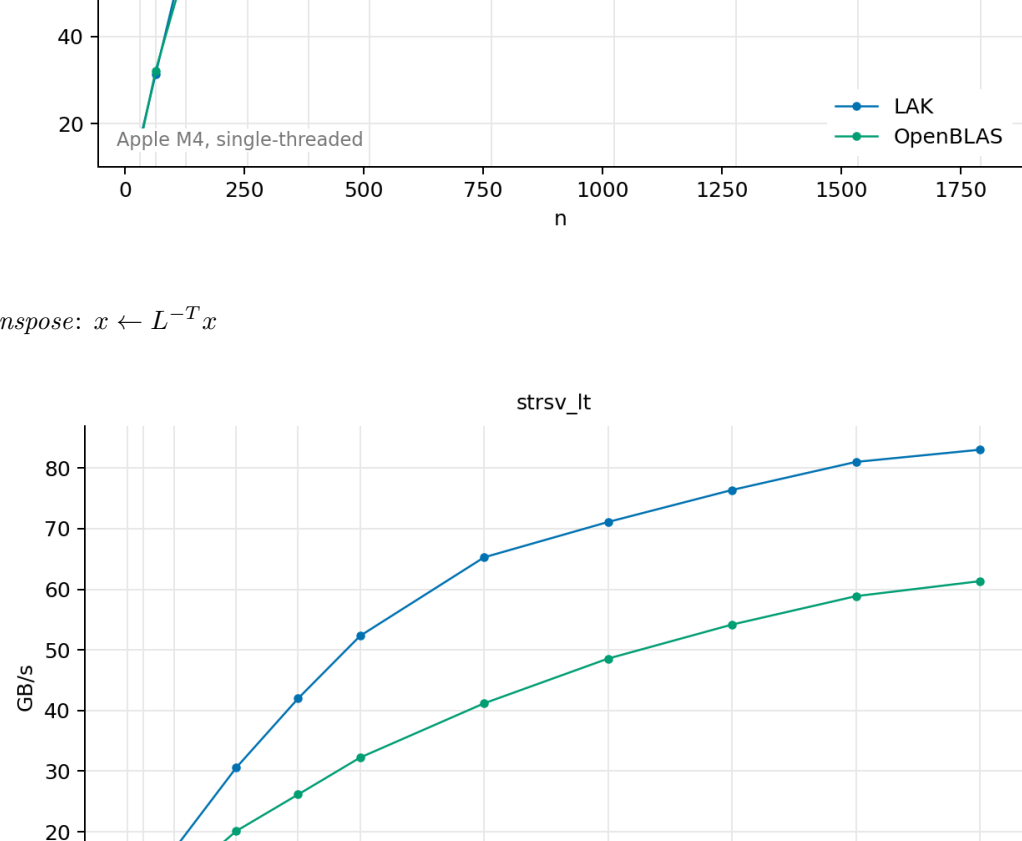
STRMV, upper transpose: $x \leftarrow U^T x$



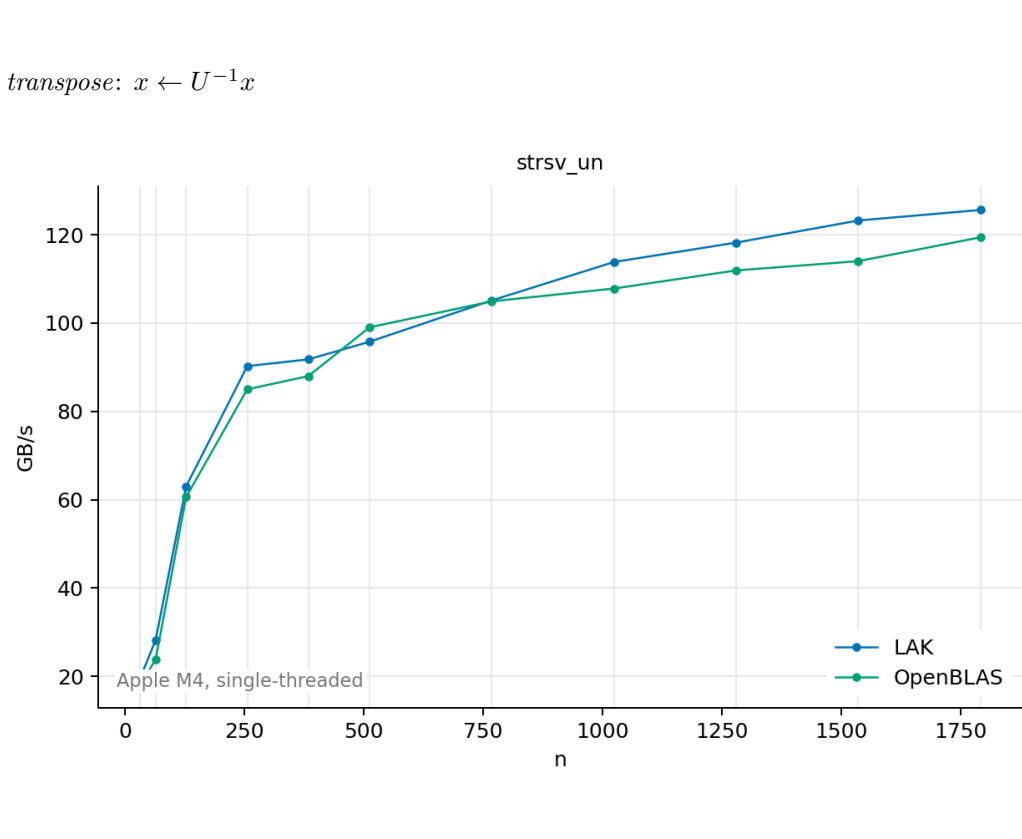
DTRMV, lower no transpose: $x \leftarrow Lx$



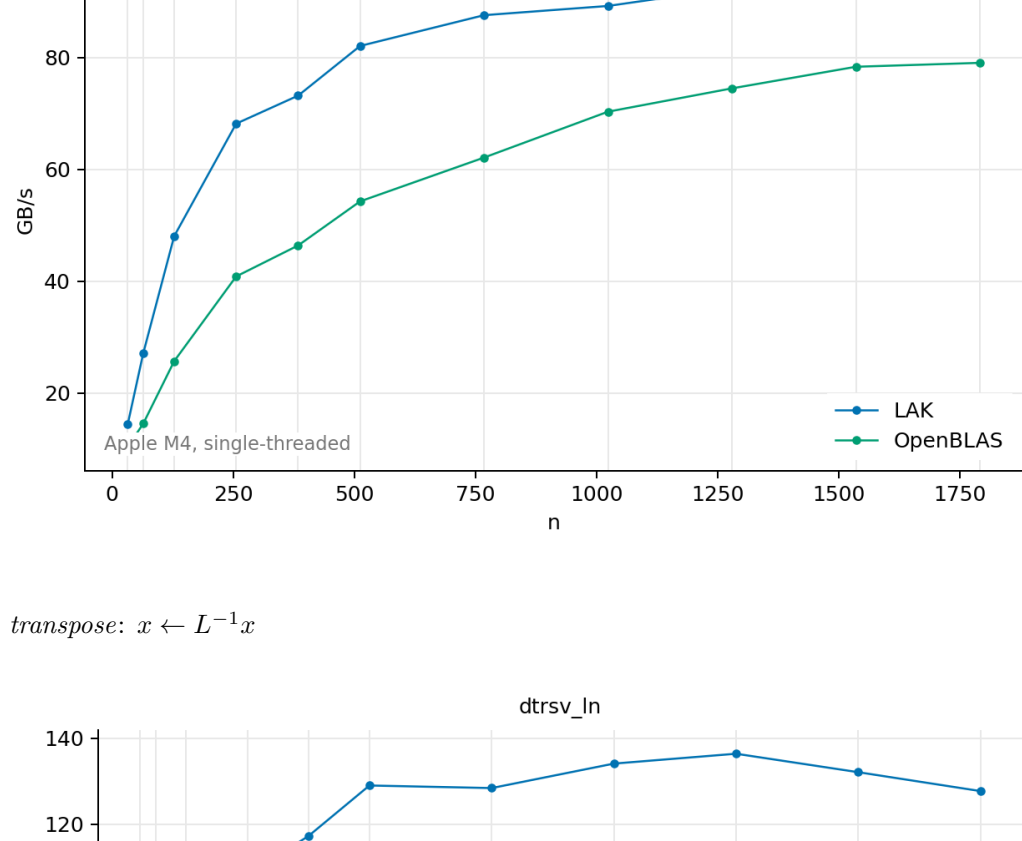
DTRMV, lower transpose: $x \leftarrow L^T x$



DTRMV, upper no transpose: $x \leftarrow Ux$

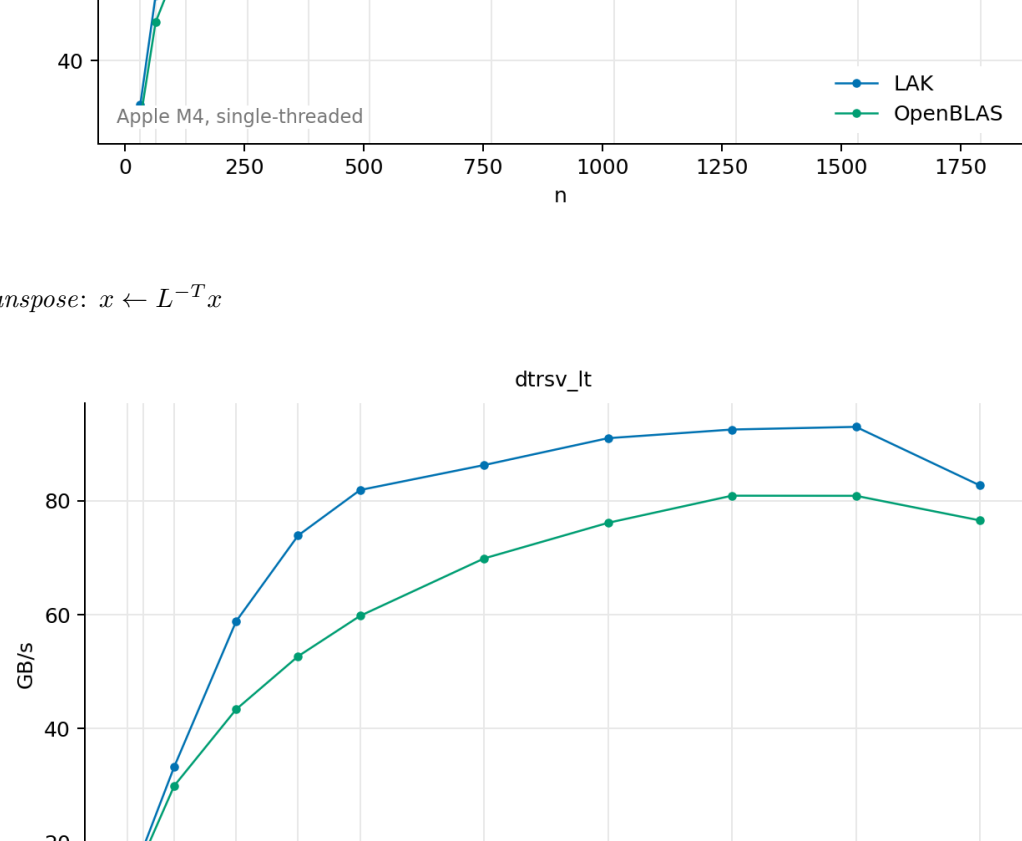


DTRMV, upper transpose: $x \leftarrow U^T x$

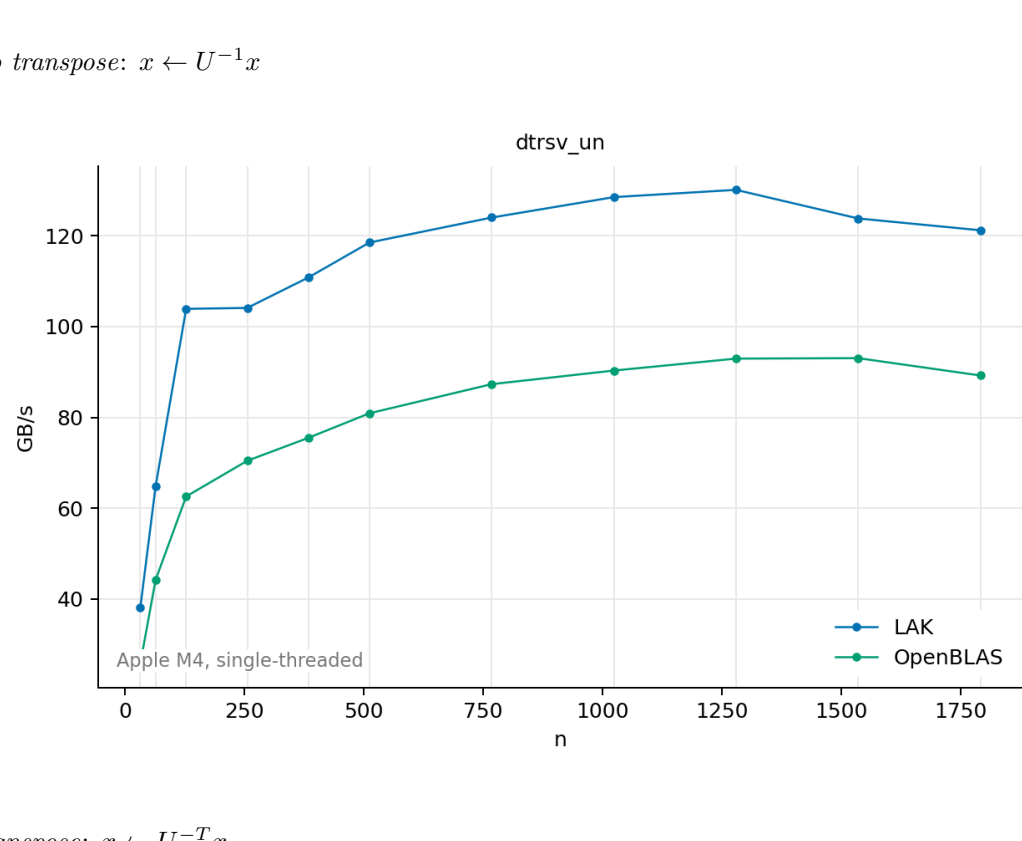


TRSV

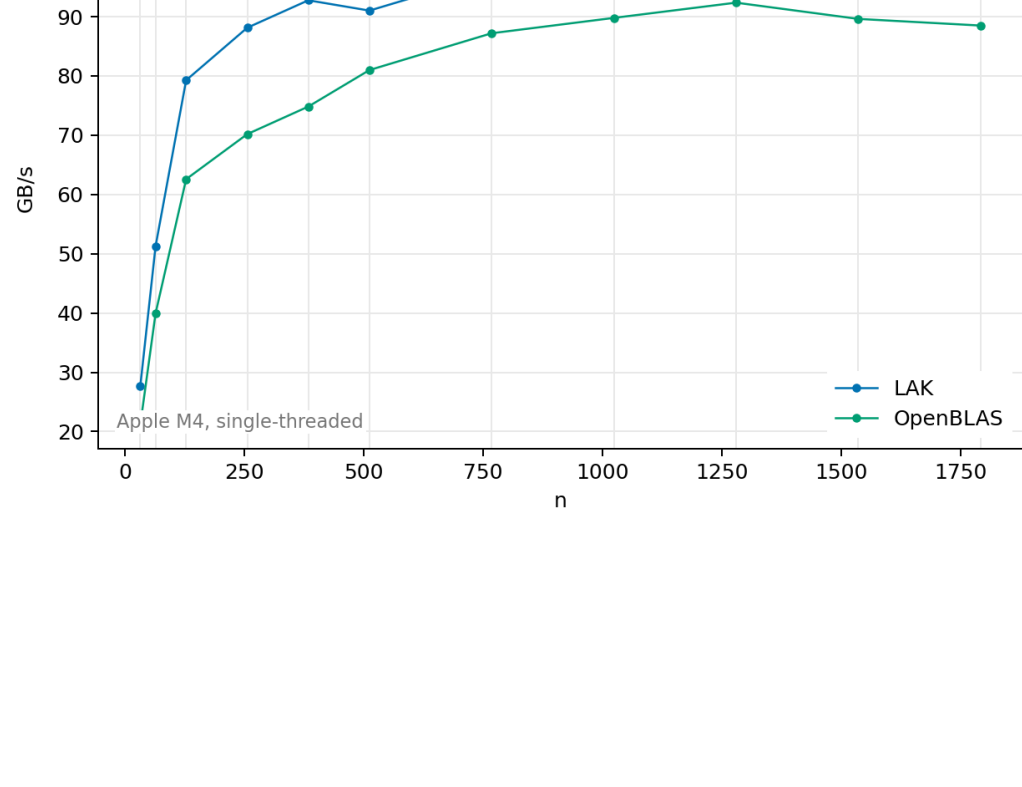
STRSV, lower no transpose: $x \leftarrow L^{-1}x$



STRSV, lower transpose: $x \leftarrow L^{-T} x$



STRSV, upper no transpose: $x \leftarrow U^{-1}x$



STRSV, upper transpose: $x \leftarrow U^{-T} x$

DTRSV, lower no transpose: $x \leftarrow L^{-1}x$

DTRSV, lower transpose: $x \leftarrow L^{-T} x$

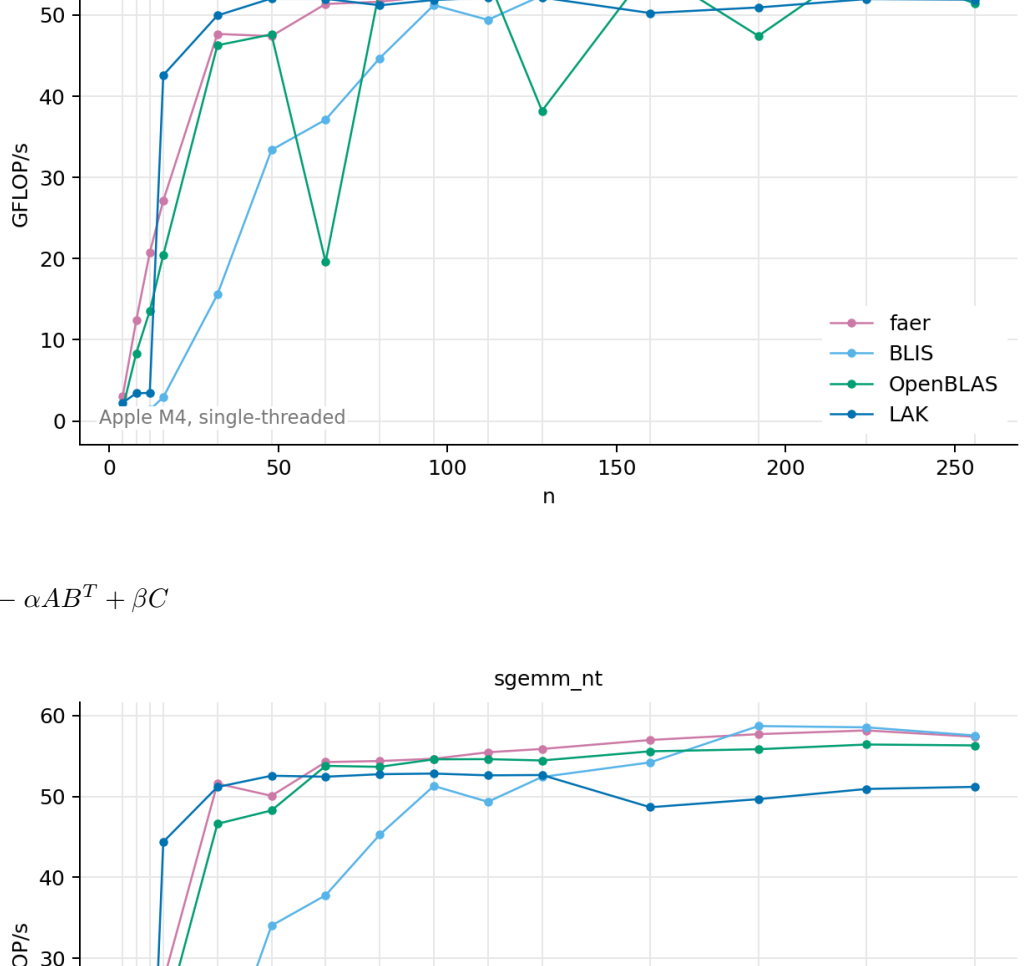
DTRSV, upper no transpose: $x \leftarrow U^{-1}x$

DTRSV, upper transpose: $x \leftarrow U^{-T} x$

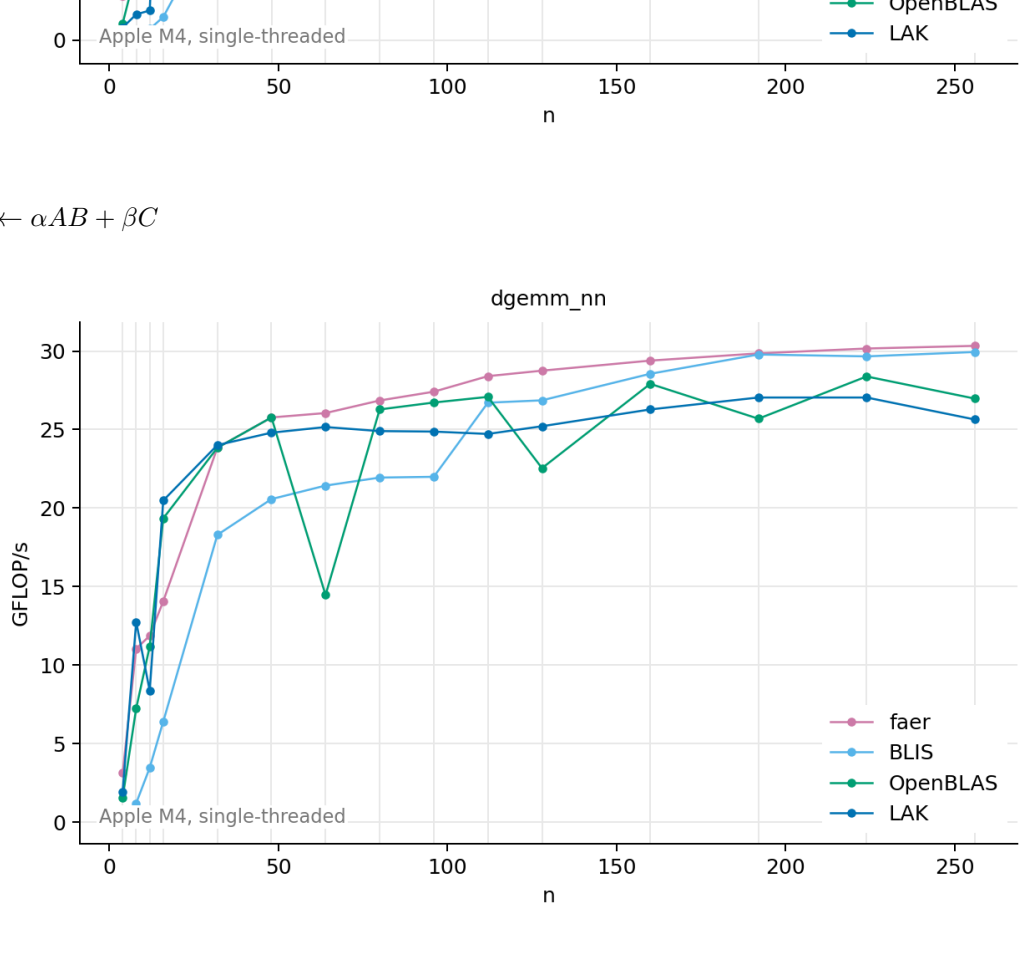
Level-3

GEMM

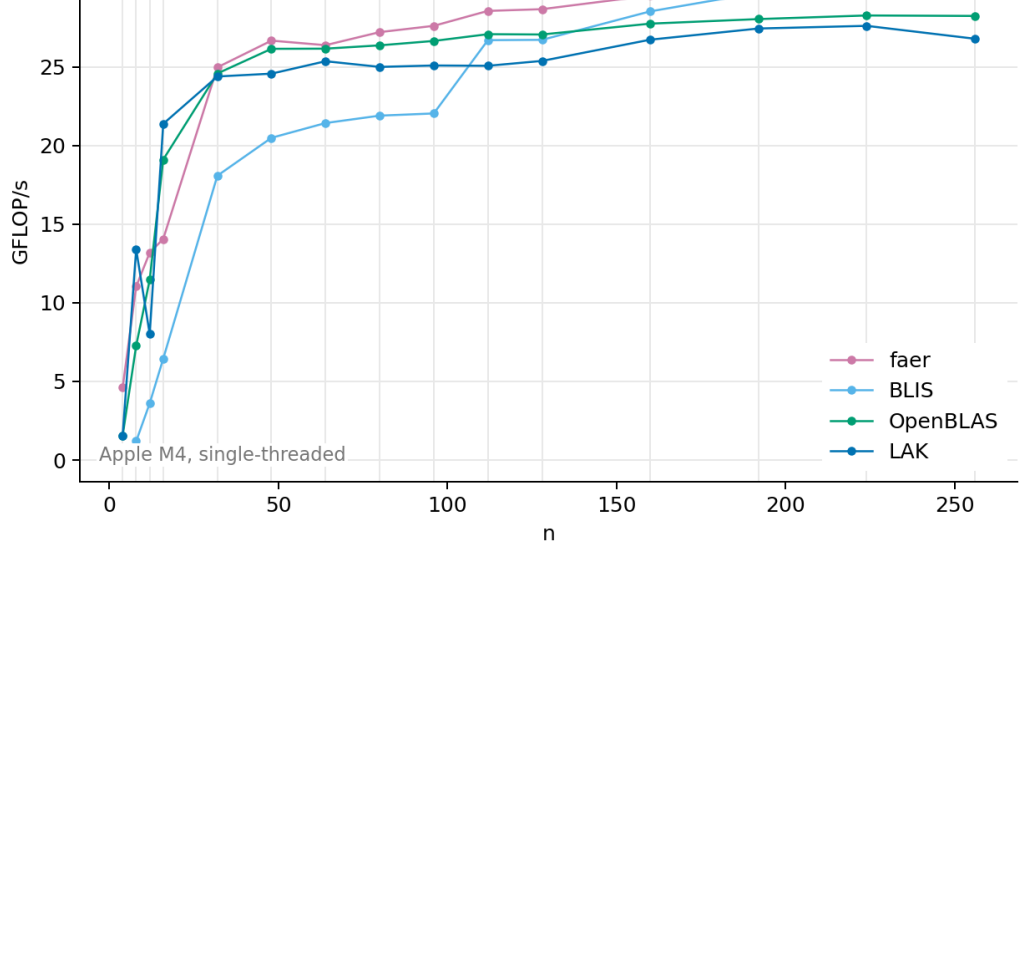
$SGEMM, NN: C \leftarrow \alpha AB + \beta C$



$SGEMM, NT: C \leftarrow \alpha AB^T + \beta C$



$DGEMM, NN: C \leftarrow \alpha AB + \beta C$



$DGEMM, NT: C \leftarrow \alpha AB^T + \beta C$

